Method for voltage-sag-source detection by investigating slope of the system trajectory

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Abstract: A method of determining the voltage-sag-source direction is proposed. The slope of a line fitting two parameters during the disturbance is employed for the detection. A least-squares method is used to perform the line fitting. The voltage-sag-source direction is determined by examining the sign of the slope. The method is verified to be promising by theoretical analysis, simulations and actual field measurements. Simplicity, ease of implementation and robustness are the advantages of the method.

1 Introduction

Major disturbances such as faults in the power system can cause voltage sags throughout a network. These sags are detrimental to those loads which are very sensitive to voltage quality. Although it is critically important to develop voltage-sag-mitigation devices to keep the disturbances as small as possible, locating the source of disturbance has to be done as a first step. This is because only after information about a disturbance location is available, can power-quality trouble-shooting, diagnosis and mitigation be carried out. Moreover, any disputes about the major responsible party can be resolved fairly, and the standard compliance can be fully accessed.

To our knowledge, however, little work has been done on voltage-sag-source location. [1] proposed a method based on the disturbance power and energy to determine on which side of a recording device the disturbance originated, but mathematical analysis was not included in the paper. Moreover, the degree of confidence in the method will decrease if the results from the disturbance power and disturbance energy do not match. The method requires a threshold for disturbance energy to achieve reliable result interpretation; as a result, the correctness of the method depends on the selection of the threshold. Another possible way of detecting voltage-sag-source is to apply the directional relay concept [2–4], where the phase difference between the voltage and current at the monitoring point is checked during the disturbance. If current lags voltage by nearly 90°, the disturbance is located in the forward direction. Otherwise, if current leads voltage by 90°, the disturbance will be assumed to be in the reverse direction. However, this method may not be directly applicable in some single-source radial systems. Consider the radial system in Fig. 1, for example; the monitored current will not lead the voltage if the system is passively reactive. Consequently, the reverse-direction fault cannot be detected properly by the classical directional-relay method. It is

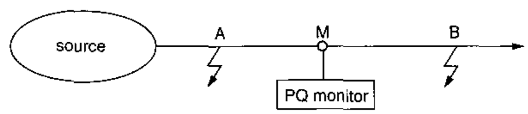

Fig. 1 *The case of a one-source system*

desirable, therefore, to find a reliable and widely effective method of determining sag-source locations in the system.

We propose a new method for determining the voltage source location. The method is based on the conclusion that the relationships between the product of voltage magnitude and power factor against current magnitude at the measurement location are not the same for different fault locations. The method first plots these two parameters during the system disturbance and then checks the slope of the line fitting the measured points. The sign of the slope indicates the direction of voltage-sag source correctly. The method is simple and easy to implement, since it depends only on the slope sign and there is no need to calculate additional parameters or set the threshold value. In addition, the method employs multiple data points; this will improve its reliability and robustness. Besides theoretical analysis, simulations on a real system and field test are performed to verify the correctness of the method in this paper; the results are promising.

2 Concepts of the proposed method

2.1 Basic idea

The proposed method can easily be understood by considering a simple case shown in Fig. 1. In this case, assume that there is one power source supplying a passive load and a power-quality monitor is placed at point M. Consider two fault cases which cause voltage sag at measurement location M: case 1 involves a three-phase short-circuit fault at point A and case 2 a similar fault at point B.

As a result of the fault, the voltage measured at M will drop as a sag; the current at M, however, will show different behaviours for these two cases:

For case 1, the measured current will be the load current at the lower voltage, which is usually smaller than the normal load current. In other words, the current measured at M will decrease during the voltage sag.

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For case 2, the measured current will be the sum of the fault current and the load current at the lower voltage, which is usually larger than the normal load current. Therefore, the current measured at M will increase during the voltage sag.

The above observations can be presented graphically in Fig. 2, where each event is characterised by two dots which represent the presag and postsag quantities (i.e. V and I before and during the sag), respectively. It can be seen that the slopes of the lines connecting the two dots are different for the two cases. A positive slope shows that the sag is from upstream and a negative slope shows that it is from downstream. This conclusion forms the basis of our idea: determine the slope of a sag disturbance and deduce the direction of sag source from the slope.

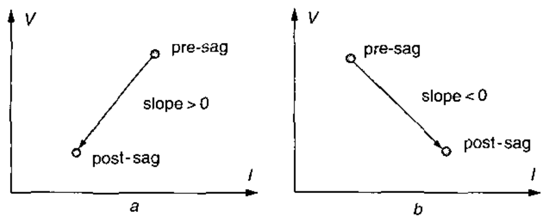

Fig. 2 *Slope characteristics of the V–I trajectory*
a Case 1 (fault at point A)
b Case 2 (fault at point B)

Although the idea was conceived for a one-source system, it has been found that it is equally valid for two-source systems.

2.2 Detailed description of the proposed method

The problem of voltage-sag-source detection is to determine on which side of the recording device the disturbance is located. As shown in Fig. 3, a monitor is installed between a particular customer and the rest of the system. The location of the disturbance used in this paper is defined with respect to the direction of the fundamental active power. Seen from the measurement location, if the fault is in the downstream direction of the active power flow, it is called a 'downstream fault', otherwise it is called an 'upstream fault'.

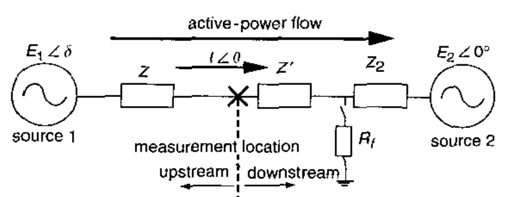

Fig. 3 *Equivalent circuit for the sag-source-detection analysis*

The proposed concept will be illustrated by studying the system in Fig. 3. A resistance (R_f) is used to model fault or heavy load switching, which are the major sources of voltage sags. The proposed method also assumes that the parameters on the healthy side do not change during the disturbance. In Fig. 3, consider a disturbance which occurs on the right-hand side of the measurement location, while the parameters on the left-hand side remain constant. We then obtain:

$$\boldsymbol{V} = \boldsymbol{E}_1 - \boldsymbol{IZ} \tag{1}$$

where $\boldsymbol{V}$ and $\boldsymbol{I}$ can be measured directly. Multiply $\boldsymbol{I}^*$ on both sides of (1) and abstract the real part; one obtains

$$VI\cos\theta_2 = E_1 I \cos\theta_1 - I^2R \tag{2}$$

where θ_2 is phase difference between $\boldsymbol{V}$ and $\boldsymbol{I}$, θ_1 is phase difference between $\boldsymbol{E}$ and $\boldsymbol{I}$, and R is the real part of $\boldsymbol{Z}$. Rewrite (2) as:

$$V\cos\theta_2 = -RI + E_1\cos\theta_1 \tag{3}$$

If $\cos\theta_2 > 0$, i.e, the active power P flows in the direction shown in Fig. 3, then the disturbance is downstream and one obtains $|V\cos\theta_2| = V\cos\theta_2$. The proposed method is inspired by the fact that each point of $(I, |V\cos\theta_2|)$ is at a certain line with slope of $-R$ (Fig. 4*a*). One can imagine that if $\cos\theta_1$ does not change greatly during the disturbance, the slope of the line fitting the points of $(I, |V\cos\theta_2|)$ during the disturbance will be negative for such a downstream event.

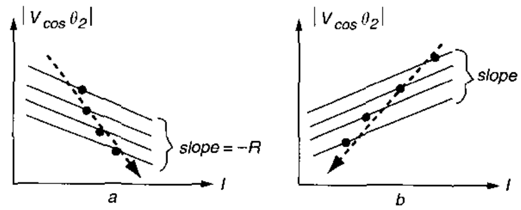

Fig. 4 *Equivalent circuit for the sag-source-detection analysis*
a Downstream event
b Upstream event

If $\cos\theta_2 < 0$, i.e. P flows from E_2 to E_1, the disturbance is in the upstream direction: one can easily obtain:

$$|V\cos\theta_2| = RI - E_1\cos\theta_1 \tag{4}$$

i.e. each point of $(I, |V\cos\theta_2|)$ is on a certain line with slope of $+R$ (Fig. 4*b*). Similarly, the line connecting points of $(I, |V\cos\theta_2|)$ during the upstream event will have a positive slope as long as $\cos\theta_1$ does not change much during the disturbance.

Further study shows that the assumption that $\cos\theta_1$ does not change much during the event cannot always be satisfied. This is because the phase of current $\boldsymbol{I}$ may change significantly during the disturbance. Fortunately, it can be proved theoretically that, during the disturbance, $|\cos\theta_1|$ will usually decrease while I will increase for a downstream event and decrease for an upstream event (see Appendix, Section 7). This will guarantee that the line connecting the points of $(I, |V\cos\theta_2|)$ will have negative slope for a downstream disturbance and positive slope for an upstream disturbance, as shown in Fig. 4. Therefore, by simply investigating the slope of the line fitting those points of $(I, |V\cos\theta_2|)$, one can reliably find the direction of the disturbance.

2.3 Applicability of the method in one-source system

To check the applicability of the method to a radial system, consider again the one-source system from which the basic idea was developed. Consider the one-source system shown in Fig. 5. If we monitor at the left-hand (or right-hand) side of fault resistor R_f, a downstream (or upstream) fault should be observed. Obviously the monitored current will increase for a downstream event and decrease for an upstream event. In addition, the voltage magnitude always decreases significantly for a voltage-sag event; the value of

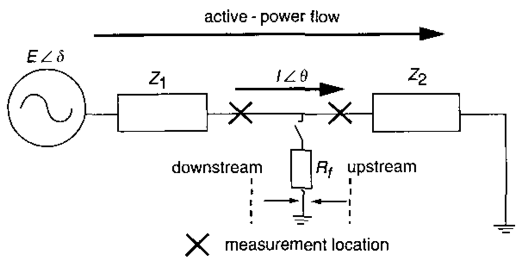

Fig. 5 *One-source system with two different monitoring points*

$\cos\theta_2$ does change for an upstream fault [$\theta_2 = \arg(\mathbf{Z}_2)$] and changes little for a downstream one [this is because $\mathbf{Z}_2$ is dominated by load impedance and, accordingly, almost resistive, $\theta_2 = \arg(\mathbf{Z}_2 \parallel R_f)$ does not change much]. As a result, if one applies a straight line to fit the points of (I, $|V\cos\theta_2|$), it will have a negative slope for a downstream disturbance and a positive slope for an upstream one. Therefore, the proposed method can apply directly to one-source radial system.

2.4 Exploiting power direction change to detect upstream event

In the above analysis, the active-power direction is assumed to be unchanged before and after the disturbance occurs. In fact, the power direction may reverse following a heavy disturbance but this is obviously only possible during an upstream disturbance. For a downstream fault, it is certain that the active-power direction remains unchanged before and during the fault. This observation can be used for the upstream-fault interpretation in this way: if the active-power direction reverses, then the disturbance is located in the upstream direction. This auxiliary rule, together with slope checking can give reliable sag-source detection results.

2.5 Summary of the implementation steps

In summary, the proposed method can easily be implemented as follows:

(i) Abstract the fundamental components from the recording voltage and current once voltage sag is captured; calculate $\cos\theta_2$ for each point.

(ii) Plot the coordinates of (I, $|V\cos\theta_2|$) during the voltage sag.

(iii) Apply the least-squares method to fit the points with a straight line.

(iv) Check the sign of the slope. If it is negative, the disturbance is located downstream, while a positive slope represents an upstream disturbance.

(v) If the active-power-direction reverse is detected during the event, the voltage-sag source is upstream.

3 Simulation and field-measurement results

The proposed method will be verified in this Section with simulation results from actual system and field measurements.

3.1 Simulation results

Firstly, a real power system is selected for simulations. Fig. 6 shows its simplified diagram. Three-phase-to-ground faults through resistance are simulated at five different

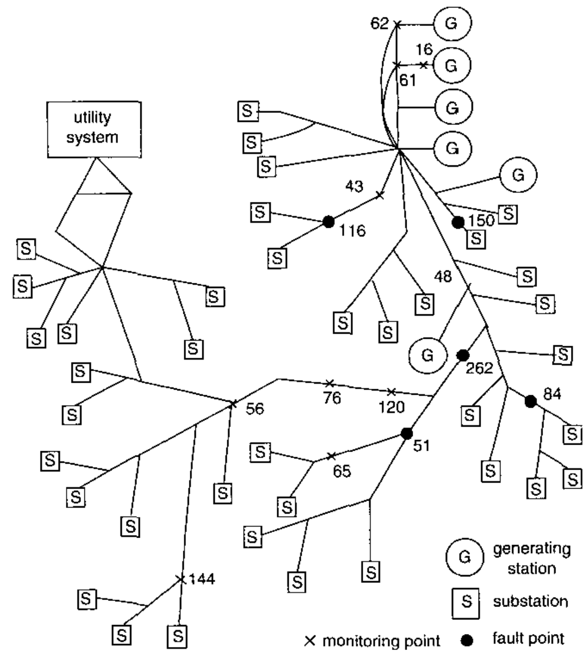

Fig. 6 *Simplified system diagram for the simulation*

locations. Ten buses which suffer voltage sags are selected to test the proposed method. Each fault will be observed at two selected buses at which the fault appears as downstream at one bus and upstream at the other. Table 1 tabulates the selected buses' number. Figs. 7 and 8 plot the trajectories of (I, $|V\cos\theta_2|$) for the five faults at downstream and upstream locations, respectively.

Table 1: Bus-selection details

Fault bus	Monitoring buses	
	Downstream fault	Upstream fault
262	61	56
150	62	76
84	43	120
51	48	144
116	16	65

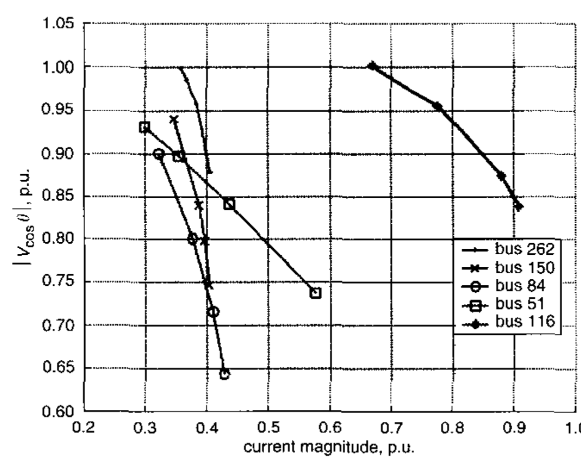

Fig. 7 *Simulation results for the downstream fault*

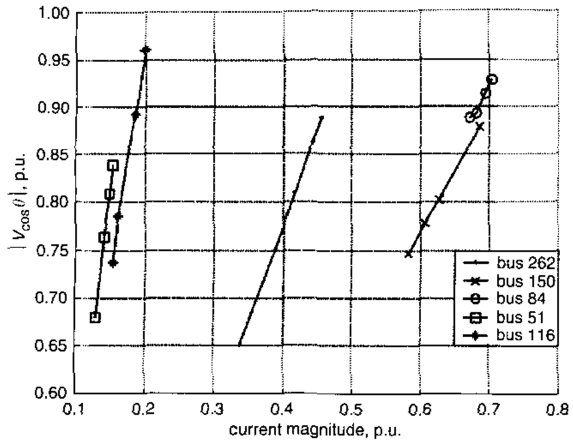

Fig. 8 *Simulation results for the upstream fault*

The simulation results also agree very well with the theoretical conclusion that the line fitting points (I, $|V\cos\theta_2|$) will yield a positive slope for the upstream fault and a negative slope for the downstream fault. The simulated voltage drop is about 20–30%; this is moderate compared with the typical 10–90% voltage drop for sags in IEEE Standard *1159-1995* [5]. Hence the simulations could be used to estimate the actual cases in a real power system.

3.2 Field measurements

The proposed method has also been applied to field measurements of one industrial plant. The measured system consists of four parts as shown in Fig. 9:

Part A: The 225 kV system is stepped down to 12.6 kV by two 50 MVA transformers. Two cables (5 km for each) are then used to feed two normally closed buses. A power-quality monitor is located at one cable terminal.

Part B: There are two groups of 600 V VFDs fed from a 12.47 kV bus; each group has a capacity of 2.7 MW. This part also has 5th- and 7th-harmonic filters and some office load.

Part C: This part mainly includes five generators (total capacity 4 MW) and six VFDs (total capacity 2.9 MW). Therefore, a two-source system is monitored in this field-measurement example.

Part D: This part is mainly commercial load, such as office-building load and heating.

During the monitoring a downstream disturbance was automatically captured and the $|V\cos\theta_2|$ and I plot is shown in Fig. 10. The disturbance is located at the downstream side, since the slope of the fitting line is negative. This agrees with the actual case.

Both the simulation and field measurement show promising results. The sign of the slope can indicate correctly whether the fault originates on the downstream or upstream side. The method is simple and easy to

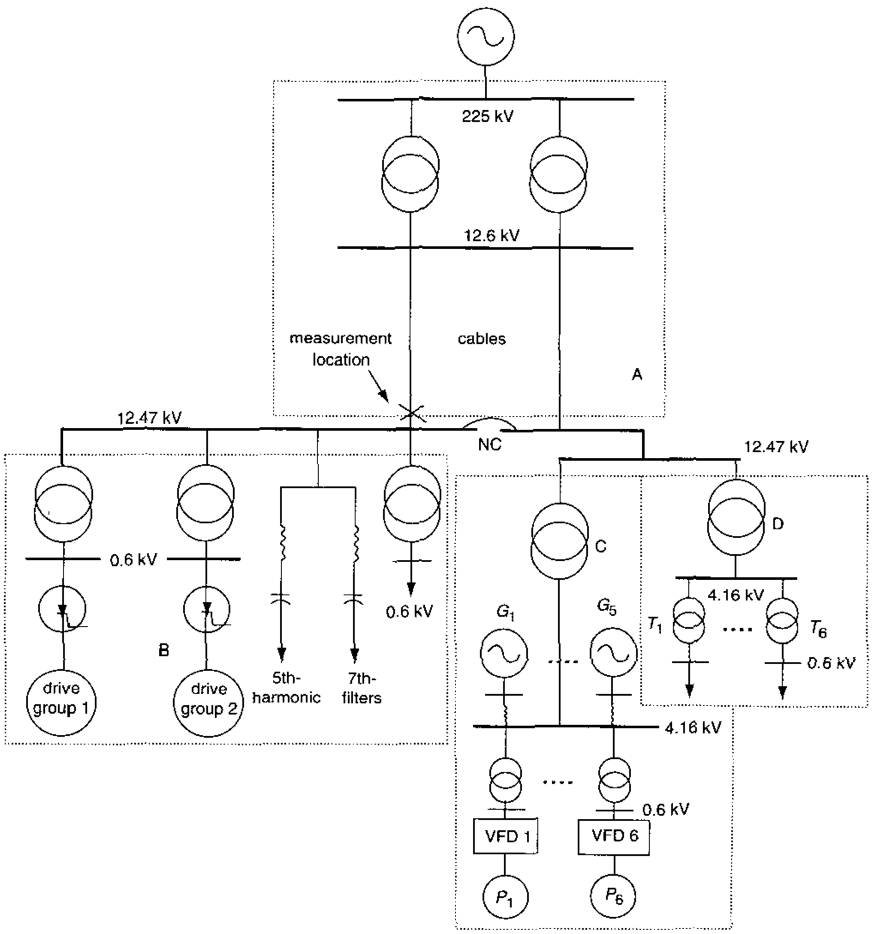

Fig. 9 *Diagram of the field measurement system*

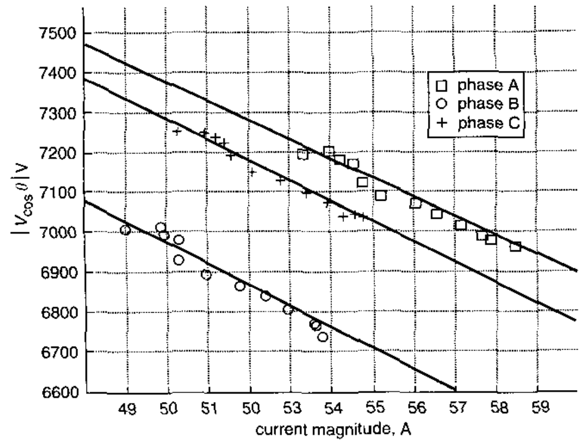

Fig. 10 *Field measurement of a downstream disturbance*

implement since $|V\cos\theta_2|$ and I can be obtained directly from the measurements. Moreover, the method does not depend entirely on one point of data and requires no threshold; these will improve its reliability and robustness.

4 Conclusions

A new method is proposed to determine the voltage-sag-source direction. It is to check the slope of a line fitting points of (I, $|V\cos\theta_2|$). Faults originated on different locations relative to the monitoring device will yield different slope signs. The main conclusions of this paper are:

(i) If the slope is negative, the fault is located downstream. If the slope is positive, the fault is located upstream. This conclusion is based on the assumption that the parameters on the healthy side do not change.

(ii) If the active power direction is reversed during the fault, it is certain that the fault is located upstream.

(iii) The proposed method utilises a least-squares-based curve-fitting technique. It does not simply rely on one-point measurement, so the result is reliable and robust.

(iv) The proposed method can be applied to both single-source and two-source systems.

(v) The proposed method is easy to implement and does not require a threshold for result interpretation. The sign of the slope is sufficient to determine the location of disturbance.

5 Acknowledgment

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6 References

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7 Appendix

In this Appendix, the current magnitude and phase changes at measurement location due to disturbances in a downstream or upstream direction are derived.

7.1 For downstream event

Refer to Fig. 3. Let $\boldsymbol{Z}_l = \boldsymbol{Z} + \boldsymbol{Z}'$ and perform Y–Δ impedance transformation; one can obtain a new equivalent circuit as shown in Fig. 11.

For simplicity assume $E_1 = E_2$ and $\boldsymbol{Z}_1 = \boldsymbol{Z}_2 = jX$, then the current delivered from $\boldsymbol{E}_1$ is

$$\boldsymbol{I} = \frac{E\angle\delta + 2E\dfrac{R}{X}\sin\dfrac{\delta}{2}\angle\dfrac{\delta}{2}}{2R_f + jX} \tag{5}$$

To find the change of I while the fault resistor applies, investigate

$$\frac{\partial I^2}{\partial R_f} = 8E^2X\sin\delta\left\{\frac{-R_f^2 - \dfrac{XR_f}{2}\cot\dfrac{\delta}{2} + \dfrac{1}{4}X^2}{(4R_f^2X + X^3)^2}\right\} \tag{6}$$

The zero of the numerator in (6) are found at

$$R_f = \frac{X}{4}\left\{\sqrt{\left(\cot^2\frac{\delta}{2} + 4\right)} - \cot\frac{\delta}{2}\right\} \tag{7}$$

That is, as long as

$$R_f > \frac{X}{4}\left\{\sqrt{\left(\cot^2\frac{\delta}{2} + 4\right)} - \cot\frac{\delta}{2}\right\},$$

$\partial I^2/\partial R_f$ will always be negative and consequently the current magnitude will increase while R_f decreases for a downstream fault. For $10° \le \delta \le 90°$, the limit of R_f given by (7) varies from $0.043X$ to $0.309X$.

The phase difference between $\boldsymbol{E}_1$ and $\boldsymbol{I}$ can be derived as

$$\theta_1 = \frac{\delta}{2} - \tan^{-1}\left(\frac{\sin\dfrac{\delta}{2}}{\cos\dfrac{\delta}{2} + 2\dfrac{R_f}{X}\sin\dfrac{\delta}{2}}\right) + \tan^{-1}\left(\frac{X}{2R_f}\right) \tag{8}$$

The change of this phase difference against R_f is

$$\frac{\partial\theta_1}{\partial R_f} = \frac{-X^2\cot^2\dfrac{\delta}{2} - 4R_fX\cot\dfrac{\delta}{2}}{(4R_f^2 + X^2)\left\{\left(X\cot\dfrac{\delta}{2} + 2R_f\right)^2 + X^2\right\}} \tag{9}$$

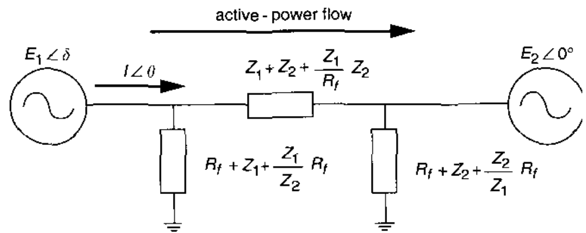

Fig. 11 *Equivalent circuit of a two-source system*

which is always negative. In summary, when R_f decreases (this can be used to simulate the development of a fault), $E\cos\theta_1$ will decrease and I will increase for the downstream fault. Refer to Fig. 4*a*: one can easily find that the line fitting the points of $(I,\ V\cos\theta_2|)$ will have a negative slope for downstream disturbance.

7.2 For upstream event

Following the similar analysis procedure, for the upstream fault it is found that as long as

$$R_f > \frac{X}{4}\left\{\sqrt{\left(\cot^2\frac{\delta}{2}+4\right)}+\cot\frac{\delta}{2}\right\},$$

$\partial I^2/\partial R_f$ will always be positive. This means the current magnitude will decrease while R_f decreases for upstream fault. It should also be borne in mind that, when $\partial I^2/\partial R_f$ tends to be negative (R_f is very small), the active-power direction will change. As discussed in Section 2.4, once the power-direction reverse is detected, one can locate the disturbance as upstream. The combination of the slope checking and power-direction monitoring can successfully identify the upstream fault.

By checking the phase difference between $\boldsymbol{E_1}$ and $\boldsymbol{I}$, it is found that, as long as

$$R_f > \frac{X}{4}\cot\frac{\delta}{2},$$

$|\cos\theta_1|$ will decrease with R_f. Note that this value is smaller than that at which $\partial I^2/\partial R_f$ changes sign. Therefore, when R_f decreases, both $|E_1\cos\theta_1|$ and current magnitude will decrease for an upstream fault. Consequently a positive slope will be obtained if the points of $(I,\ |V\cos\theta_2|)$ are fitted to a straight line.

7.3 Conclusions

(i) For a downstream fault, I increases and $|E_1\cos\theta_1|$ decreases with the development of fault.

(ii) For an upstream fault, when the fault resistor is greater than a threshold, I and $|E_1\cos\theta_1|$ decrease with the development of the fault. When R_f is below the threshold, this conclusion reverses, but the power direction may by then have changed. This is helpful in detecting the upstream fault. Thus the slope method and power-direction check can be combined correctly to detect the upstream fault.